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- (b) Briefly discuss the accessory respiratory organs in fishes. (4)
6. Give a comparative account of brain in vertebrates with labelled diagrams. (12)
7. Write short notes on *any three* of the following. (12)
- (i) Jaw suspensorium in vertebrates
 - (ii) Dentition in mammals
 - (iii) Types of feathers
 - (iv) Autonomic nervous system
 - (v) Scales in fishes

(500)

[This question paper contains 4 printed pages.]

29.05.2024 (M)
Your Roll No.

Sr. No. of Question Paper : 2956

Unique Paper Code : 32231401

Name of the Paper : Comparative Anatomy of Vertebrates

Name of the Course : B.Sc. Hons. Zoology

Semester : IV - LOCF

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Answer any **five** questions in all, including question 1 which is compulsory.
3. Draw well-labelled diagrams wherever necessary.

1. (a) Define the following terms : (5)

- (i) Iter
- (ii) Opisthonephros
- (iii) Plastron

P.T.O.

(iv) Synsacrum

(v) Holobranch

(b) Differentiate between the following terms : (10)

(i) Monocondylic and dicondylic skull

(ii) External and internal glomeruli

(iii) Spinal and cranial nerves

(iv) Nail and claw

(v) Parotid and paratoid gland

(c) State whether the following statement is true or false. (4)

(i) Syrinx is the voice box of mammals.

(ii) The neurocrania of chondrichthyans are well ossified.

(iii) Cycloid scales are epidermal derivatives.

(iv) In adult birds only the right aortic arch is functional.

(d) Give the exact location and function of the following. (8)

(i) Meibomian gland

(ii) Fovea centralis

(iii) Foramen Panizza

(iv) Foramen magnum

2. With the help of labelled diagrams, illustrate the evolution of aortic arches in vertebrates. (12)

3. (a) Describe the evolution of male urinogenital ducts in vertebrates. (6)

(b) Comment upon the ruminant stomach. (6)

4. (a) Classify and give functions of various types of receptors found in vertebrates. (7)

(b) Describe the types of centrum in vertebrates. (5)

5. (a) Explain the structure and functioning of the gills in cartilaginous and bony fishes. (8)